IONIC EQUILIBRIUM, ACID & BASE, pH, BUFFER & INDICATOR

1. Solution is acidic when:

(1) $[H_3O^+] = [OH^-]$ (2) $[H_3O^+] < [OH^-]$

(3) $[H_3O^+] > [OH^-]$ (4) None

2. The pH of 0.2 M mono basic acid is 3.50 then K_a is:

(1) 3.2×10^{-8} (2) 5.12×10^{-7}

(3) 5.12×10^{-8} (4) None

3. The pH of 0.004 M nicotine solution is 9.7 then pK_b is–

(1) 6.20 (2) 4.20

(3) 9.71 (4) 5.19

4. Among these which is strongest base–

(1) H_2A (2) HA^-

(3) A^{-2} (4) $H_3A^{\oplus}$

5. Correct order of acidic strength is–

(1) $CH_4 < H_2O < NH_3 < HF$

(2) $H_2O < NH_3 < CH_4 < HF$

(3) $CH_4 < NH_3 < H_2O < HF$

(4) $HF < NH_3 < H_2O < CH_4$

6. Correct order of basic strength is–

(1) $OH^- > HSe^- > HS^- > HTe^-$

(2) $HTe^- > HSe^- > SH^- > OH^-$

(3) $OH^- > HS^- > HSe^- > HTe^-$

(4) $HS^- > HTe^- > OH^- > HSe^-$

7. Which pair shows common ion effect?

(1) HCl + NaCl (2) HCl + KOH

(3) HCN + KCN (4) $Ba(NO_3)_2 + BaCl_2$

8. pH of solution when 50 mL of 0.10 M ammonia solution is treated with 50 mL of 0.05 M HCl solution–

(pK_b of ammonia = 4.74)

(1) 8.26 (2) 9.26

(3) 4.74 (4) None

9. For WA WB type of salt, pH is given by this formula

$$pH = 7 + \frac{1}{2}(pK_a - pK_b)$$

The pH of solution can be greater than 7, if the difference between pK_a and pK_b is –

(1) Negative (2) zero

(3) positive (4) None

10. The solubility of salts depends on which factor (s)–

(1) Temperature

(2) lattice enthalpy of salt

(3) Solvation enthalpy of ions in a solution

(4) All of these

11. If the molar solubility of Zirconium phosphate is S then solubility product (K_{sp}) of zironium phosphate is–

(1) $108\ S^5$ (2) $729\ S^6$

(3) $6912\ S^7$ (4) None

12. Which salt is most soluble?

(1) CaF_2 ($K_{sp} = 5.3 \times 10^{-9}$)

(2) $BaCrO_4$ ($K_{sp} = 1.2 \times 10^{-10}$)

(3) CaC_2O_4 ($K_{sp} = 4 \times 10^{-9}$)

(4) Hg_2Cl_2 ($K_{sp} = 1.3 \times 10^{-18}$)

13. The solubility of salts of weak acids increases at

(1) higher pH (2) Lower pH

(3) same pH (4) Can't say anything

14. Which is lewis acid?

(1) H_2O (2) NH_4^+

(3) H^+ (4) OH^-

15. Molar solubility of $Cd(OH)_2$ ($K_{sp} = 2.5 \times 10^{-14}$) in 0.1 M KOH solution is–

(1) 2.5×10^{-7} M (2) 5×10^{-8} M

(3) 5×10^{-7} M (4) 2.5×10^{-12} M

16. Molar solubility of $Ca(OH)_2$ in a solution that has a pH of 12. [K_{sp} $[Ca(OH)_2] = 5.6 \times 10^{-12}$]

(1) 5.6×10^{-10} M (2) 5.6×10^{-8} M

(3) 4×10^{-4} M (4) None

17. Degree of hydrolysis of the following is independent of concentration for :

I. NH_4CN II. NH_4HCO_3

III. Mg_2S IV. CH_3NH_3Cl

(1) I, II, III, IV (2) I, IV

(3) I, III, IV (4) I, II, III

18. What is $\left[NH_4^+\right]$ in a solution in a mixture of 0.02 M NH_3 and 0.01 M KOH?

[K_b (NH_3) = 1.8×10^{-5}]

(1) 3.6×10^{-5} M (2) 1.8×10^{-5} M

(3) 0.9×10^{-5} M (4) 7.2×10^{-5} M

19. The conjugate base of NH_2^- is–

(1) NH_3 (2) NH_4^+

(3) NH^{-2} (4) N_2H_4

20. pK_b of a base is 6, then K_a of its conjugate acid will be –

(1) 10^{-6} (2) 10^{-8} (3) 10^{-14} (4) 10^{-12}

21. Which of the following is correct for the dissociation of H_2CO_3 –

(1) $K_{a_1} = K_{a_2}$ (2) $K_{a_1} < K_{a_2}$

(3) $K_{a_1} > K_{a_2}$ (4) None

(Here K_{a_1} & K_{a_2} are first and second dissociation constant of H_2CO_3 respectively)

22. Aqueous solution of $Al_2(SO_4)_3$ is –

(1) Strong acidic (2) Weak acidic

(3) Strong basic (4) Weak basic

23. Which has least possibility to be a Lewis acid –

(1) CN^- (2) $SnCl_2$ (3) PCl_3 (4) I^+

24. Which of the following has maximum pH ?

(1) 1M H_2S (2) 1M H_2O

(3) 1M H_2Se (4) 1M H_2Te

25. Which of the following is strongest base in aqueous solution ?

(1) NaOH (2) RbOH (3) KOH (4) CsOH

26. On addition of KF in the solution of HF –

(1) dissociation of HF increases.

(2) concentration of H^+ ions increases.

(3) concentration of H^+ ions decreases.

(4) concentration of F^- ions decreaseses.

27. The aqueous solution of $FeCl_3$ is acidic because –

(1) Fe^{+3} ions react with water to give H^+ ions in solution.

(2) Cl^- ions react with water to give H^+ ions in solution.

(3) It is a salt of strong base and weak acid.

(4) None of these

28. Which of the following salt solution is almost neutral and can also act as buffer solution ?

(1) NaCl (2) CH_3COONH_4

(3) NH_4Cl (4) CH_3COONa

29. The K_{sp} of $Ni(OH)_2$ is 2×10^{-15}. Its solubility in 0.1M NaOH is –

(1) 1.41×10^{-8} M (2) 2×10^{-13} M

(3) 2×10^{-14} M (4) 1.41×10^{-6} M

30. Which of the following has maximum conductivity?

(1) HF (2) HCOOH

(3) HBr (4) HCN

31. Which of the following can act as a Bronsted acid but not as an Arrhenius acid?

(1) HCl (2) HNO_2

(3) CH_3COOH (4) H_3O^+

32. The pH of 0.004 M N_2H_4 solution is 9.7. Calculate its K_b –

(1) 6.25×10^{-7} (2) 12.2×10^{-8}

(3) 1.7×10^{-6} (4) 1.5×10^{-10}

33. $pK_a + pK_b = pK_w$ is applicable for –

(1) HNO_3, NO_2^- (2) NH_4^+, NH_4Cl

(3) HCl, CN^- (4) NH_3, NH_2^-

34. The conjugate acid of CH_3COOH is –

(1) CH_3COO^- (2) $CH_3CO\overset{+}{O}H_2$

(3) CH_3COO^+ (4) H^+

35. The pH of 0.003 M HBr solution is –

(1) 2.3 (2) 3.3 (3) 3.7 (4) 2.52

36. 0.56 g KOH is dissolved in 200 ml solution. What is the pH of solution ?

(1) 1.3 (2) 12.7 (3) 2.3 (4) 11.7

37. The pH of 0.1M solution of HCNO is 2.3. The dissociation constant of the acid is –

(1) 5×10^{-5} (2) 2×10^{-3}

(3) 2.5×10^{-4} (4) 25×10^{-6}

38. The K_w of water at 310 K is 2.7×10^{-14}. The pH of water at this temperature is –

(1) 14 (2) 7 (3) 6.79 (4) 8.21

39. Which of the following is amphiprotic –

(1) $H_2PO_2^-$ (2) HPO_3^{-2} (3) HPO_4^{-2} (4) All